

Hybrid Feature-Based Detection of AI-Generated Text

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Abstract

1 Introduction

For this project, we wish to explore which linguistic or statistical features are most effective for distinguishing human- vs. AI-generated text. The list of features is not fully determined, but some relevant features might be: sentence structure- and complexity, punctuation and formatting, lexical diversity, perplexity and burstiness. Recent surveys indicate that hybrid approaches, combining statistical and neural methods, often outperform single-modality detectors in robustness and generalization (??). Thus, we will evaluate individual features as well as combinations and ensembles to identify optimal hybrid configurations.

RQ: Which linguistic or statistical features are most effective in distinguishing human- vs. AI-generated text?

2 Related Work

3 Data

To ensure our model generalizes across diverse English language sources, domains, and LLMs, and to mitigate overfitting to any single dataset's biases, we will utilize multiple datasets for training and evaluation. This approach allows for robust testing of features and hybrids in varied contexts, such as essays, Q&A, and long-form texts from different LLMs.

- AI Text Detection Pile (Hugging Face): ~1.3M samples, diverse texts (Reddit, Wikipedia, books) (?).
- AH&AITD (Arslan's Human and AI Text Database): ~12k samples, texts across domains (articles, abstracts, stories, news, reviews) (?).
- LLM - Detect AI Generated Text Dataset (Kaggle sunilthite): >28k essays (?).
- LLM - Detect AI Generated Text (Kaggle Competition): ~10k essays (?).

Dataset	Samples	Human (%)	AI (%)
AI Text Detection Pile	1.39M	73.8	26.2
HC3	85K	68.5	31.5
sunilthite	29K	60.1	39.9
DAIGT V2	45K	61.0	39.0
LLM Detect Competition	1.4K	99.8	0.2
AH&AITD	12K	50.0	50.0

Table 1: Dataset size and class distributions

- DAIGT V2 Train Dataset (Kaggle): 48k samples, essays (?).
- Human ChatGPT Comparison Corpus (HC3): ~24k rows, Q&A from domains (finance, medicine, law, open) (?).

3.1 Dataset Balancing

The datasets show varying degrees of class imbalance between the target classes human-written (0) and AI-generated (1). With the extremes being the LLM Detect Competition being 99.8% human-written, while AH&AITD is perfectly balanced. The dataset's class distribution is shown in Table ??.

The dataset was balanced using a per-source strategy, where each source contributes 50% human and 50% AI-generated samples according to its configured proportion of the total dataset size.

4 Method

First the chosen features will be extracted for each entry in the datasets. Examples would be using a language model to calculate perplexity, a tokenizer to calculate lexical diversity or a POS tagger to calculate POS tag ratios. Then, a (lightweight) classifier will be trained on these features; so far logistic regression and XGBoost are candidates. To assess individual and combined efficacy, we will test features separately and explore ensembles (e.g., stacking or voting across feature subsets or classifiers). Lastly, the feature importances will be analyzed.

4.1 Feature Extraction

For each entry in the dataset, the following features were extracted:

4.1.1 Punctuation Marks

The ratio of punctuation marks to word count was calculated for the following punctuation marks: comma, period, exclamation mark, and question mark.

4.1.2 POS-tags

The POS-tags were extracted using the spaCy library. Coarse POS tags (using

`token.pos_`

) were used, instead of fine grained tags (

`token.tag_`

), to capture the overall patterns in word class usage. Fine grained tags were excluded as they might be too specific for the objective, and yield sparse data.

4.2 Evaluation

The classifier(s) will be evaluated using accuracy, F1-score and AUROC. In addition, precision at different false-positive rates may be reported to mimic realistic screening use.

Depending on the classifier, feature importance will be evaluated using coefficients (for logistic regression), SHAP values (?) and/or permutation importance. Hybrid combinations will be benchmarked against baselines to quantify improvements in performance and robustness.

5 Experiment and Results

6 Discussion and Conclusion

A Example appendix